DATA WRANGLING

B.Tech

CSEDS 311

Report on Near-Earth Asteroid Characterization and Potentially Hazardous Asteroid (PHA) Classification

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1. Objective

A potentially hazardous asteroid (PHA) is, by NASA's definition, a near-Earth asteroid with absolute magnitude $H \leq 22$ (roughly larger than 140 m) and a minimum orbit intersection distance (MOID) of ≤ 0.05 au from Earth's orbit. The flag is therefore a definition, not a physical measurement of danger.

Research question. How effectively can an asteroid's designated hazardous status be predicted from its observable kinematic close-approach behaviour and physical properties, and what trade-offs emerge when the directly defining threshold variables (H and MOID) are left out?

The project takes three NASA/JPL sources through the full wrangling pipeline (extraction, cleaning with a log, transformation, SQLite loading with validation), derives five insights, trains and compares classifiers under a leakage-aware design, and prepares the outputs for a three-page Power BI dashboard.

2. Dataset description

Three public sources were combined. They use different identifiers and formats, so joining them is a genuine wrangling task, and every table arrives with numbers stored as text.

Source	Content	Raw rows	Columns	Raw-data issues found
CAD (JPL close-approach API)	One row per predicted/observed Earth approach within 0.05 au, 1900-2100, requested one decade at a time	43,614	14	All 14 columns are text; 14.0% of cells missing (diameter 97.6% empty); padded names; month-name dates
Sentry (JPL impact-risk table)	One row per object on the impact-monitoring list	2,222	14	Numeric fields as text; impact window stored as '2064-2064'
NeoWs (NASA catalogue browse)	Orbital elements, size estimate and the PHA label; nested JSON flattened to one row per asteroid	16,000	26	Elements as strings in nested JSON; 1.0% missing MOID; same object on several pages / under two keys

Joins depend on a designation (e.g. '433' vs '(A898 PA)') that is written differently in each source, the label's two defining variables disagree with each other in a few records, and the ML-ready table needs one row per asteroid while CAD has one row per event.

3. Methodology by pipeline stage

3.1 Data extraction

- **CAD:** 20 requests (one per decade) to the JPL close-approach API with a 0.05 au cap and diameters requested;
- **Sentry:** one request;
- **NeoWs:** the browse endpoint, 800 randomly sampled pages of 20 objects (seed 42) out of a 62,746-object catalogue, giving 16,000 records (23.5% sample after de-duplication).
- **Constraints.** NeoWs needs a NASA API key and the shared DEMO_KEY is too restricted for hundreds of pages, so a registered key is used. The downloader retries with back-off on HTTP 429/5xx, pauses 0.4 s between calls, and caches every raw response as gzip so a re-run resumes instead of re-downloading (690 network calls in the logged run).
- **Licensing.** NASA/JPL data are free and public; sources are cited. A manifest (timestamp, row counts, pages fetched) is saved so every number in this report maps to a specific snapshot.

3.2 Data cleaning

The policy: drop only rows that are unusable or exact duplicates; flag suspicious but possibly genuine values (outliers in asteroid data are often real); impute only inside the ML pipeline, fitted on training data, so the stored dataset keeps its honest gaps. A 37-step log is kept in the notebook and exported as CSV, Markdown and a database table; the decisions that mattered most are below.

Source	Issue found	Count	Action	Why
CAD	Numeric fields, dates and padded names stored as text	10 cols / 43,614 rows	Convert	Month-name parsing is locale-proof; dates cross-checked against Julian date (0 failed)
CAD	Rows missing key fields; duplicate approaches	0 / 0	Drop (none found)	Checks kept in the pipeline so a fresh pull is protected
NeoWs	Two catalogue records sharing one canonical key	1,223	Drop, keep first	A one-to-one join is required; 16,000 → 14,776 rows
NeoWs	Missing or inconsistent designation (join key)	1	Derive	Key rebuilt from the name field ('433 Eros (A898 PA)' → '433')
NeoWs	PHA label disagrees with rule (H ≤ 22 and MOID ≤ 0.05)	23	Flag, keep label	Label and elements can come from different orbit solutions; relabelling would manufacture the answer (99.84% agreement)
NeoWs	Missing MOID / missing H	156 / 3	Keep as NaN	MOID is used only to audit the label; H imputed later, training split only
NeoWs	H outside 5-35	56	Flag	Kept for review; extreme values need a physical explanation
Cross-source	NeoWs Sentry flag differs from live Sentry table	519	Flag, use Sentry API	The two snapshots drift as objects are added and removed
ML	Missing features; skewed counts; 6.4% positives	8,168 rows	In-pipeline	Median imputation + missing-indicators, log1p, scaling, class weights; fitted on training folds only

3.3 Data transformation

- **Aggregation / recency:** 43,614 CAD events grouped to 19,846 asteroids (count, first and last approach, minimum and mean miss distance, mean/max/variance of velocity). Variance is set to 0 for the 11,546 objects with a single approach.
- **Merge:** NeoWs left-joined to the aggregate and Sentry flag with a one-to-one validation, so a bad key fails loudly. 8,168 asteroids have no approach within 0.05 au; their count is a real 0 and their distance statistics stay NaN.
- **Pivot:** approaches per decade by orbit class (20 decades × 4 classes), reshaped to long format for the dashboard.

- **Derived features:** approaches_per_decade (count divided by the object's own approach span, floored at one decade), is_sentry_monitored, diameter_mean_km, size_class, speed_class and distance in lunar distances. Fact tables keep only asteroids present in the sample (29,041 CAD events and 1,439 Sentry objects excluded) so foreign keys stay valid.

3.4 Loading and storage

The result is a SQLite star schema: dim_asteroid (14,776 rows), fact_close_approach (14,573) and fact_sentry (783), with primary keys, foreign keys, CHECK constraints and indexes, plus the decade pivot, ML output tables and the cleaning log. The fact table is also exported as a per-decade partitioned CSV set (20 partitions) and flat CSVs for Power BI. Eleven SQL validation checks all pass: row counts match the dataframes, IDs are unique, no orphan foreign keys, no non-positive distance or speed, approach_count equals the fact-row count per asteroid, and the target is non-null 0/1.

4. Insights

Each insight is computed from the data; the narrative text in the notebook is generated from the same numbers.

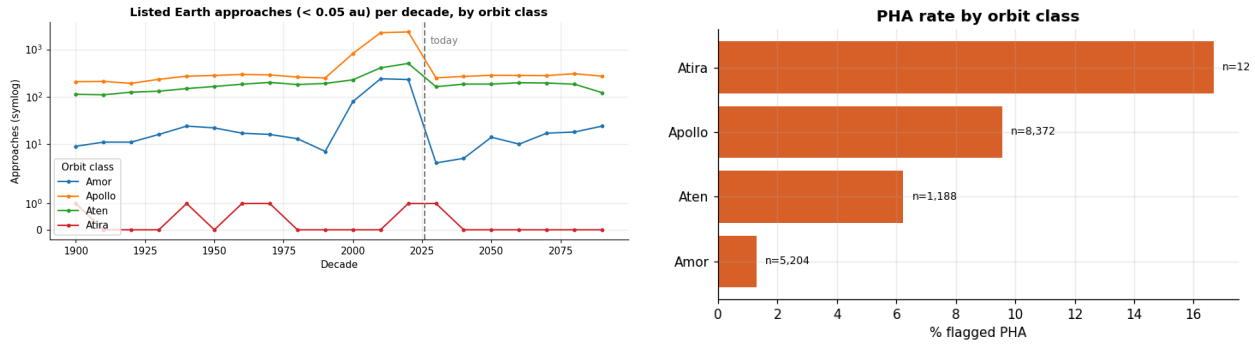


Figure 1. Listed Earth approaches per decade by orbit class (left); PHA rate by orbit class (right).

- **1. Listed approaches rise 5.3× in recent decades.** The 2000-2019 decades average 2,016 listed approaches versus 381 before 1960. Encounter physics does not change that way; it reflects discovery history and projected passes, so count-based features partly measure how well an object is observed.
- **2. Size is necessary but far from sufficient.** 21.3% of the 4,368 objects with H $\leq$ 22 are flagged hazardous, against 0.1% of 10,405 fainter objects (14 label inconsistencies, already in the log). The orbit must also come within 0.05 au.
- **3. The 'kinematic' features are not independent of MOID.** A real approach cannot be closer than the orbit-intersection distance, so minimum miss distance bounds MOID. Of 6,608 objects with both values, only 8.1% fall below the diagonal in Figure 2. This motivated the velocity-only ablation in Section 5.
- **4. PHA and Sentry answer different questions.** Only 0.3% of PHAs are on the Sentry list and 0.4% of Sentry objects are PHAs. Sentry tracks computed impact probability for individual orbits; PHA is a size-and-geometry screen. Median approach speed is 15.2 km/s for PHAs against 9.6 km/s for others.
- **5. Hazard concentrates in orbit families.** Atira objects have the highest rate (16.7%, but only 2 of 12) and Amor the lowest (1.3%). Orbit class follows from perihelion/aphelion geometry, which also drives MOID.

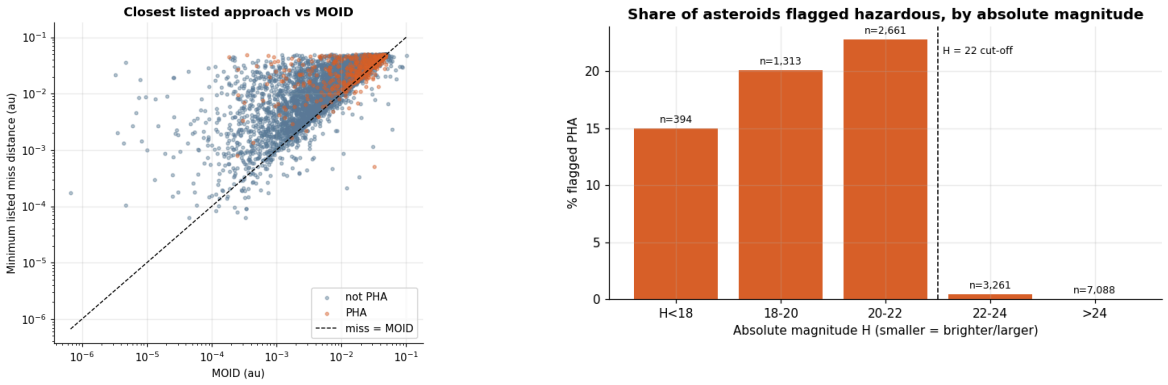


Figure 2. Closest listed approach vs MOID (left); share flagged hazardous by H band (right).

5. Machine learning

5.1 Design

Task: binary classification of the NeoWs PHA flag, one row per asteroid, so approaches of the same object can never straddle train and test. Stratified 80/20 split (11,820 train / 2,956 test; 189 test positives) with a fixed seed. Imputation, log1p and scaling sit inside a scikit-learn Pipeline, so they are fitted on training data only; 5-fold stratified cross-validation runs on the training split.

Feature sets run a leakage experiment that answers the research question directly:

- **A, Benchmark:** H, diameter min/max, minimum miss distance, mean velocity. Diameter is computed from H, so A largely contains the PHA rule.
- **B, Kinematic:** approach count, approaches per decade, mean/max/variance of velocity, minimum and mean miss distance, Sentry flag. No H or diameter.
- **C, Velocity + Sentry (ablation):** B without any distance-type feature, since Insight 3 shows miss distance bounds MOID.

Models. A stratified and a most-frequent DummyClassifier (floor), Logistic Regression (interpretable linear model) and Random Forest (non-linear interactions, built-in importances), both with balanced class weights, plus the non-learning reference rule $H \leq 22$. The better real model per feature set is chosen by cross-validated F1, so a flag-everything model cannot win on recall. Recall is the headline metric: a second operating point tunes the threshold on out-of-fold training predictions to reach 90% recall and is then applied once to the test set. Accuracy is not reported as a headline because 'predict safe' scores about 94%.

5.2 Results on the held-out test set

Feature set (model)	Operating point	Recall	Precision	F1	PR-AUC	ROC-AUC	Hits / false alarms
Dummy (stratified)	0.50	0.048	0.044	0.046	0.063	0.489	9 / 195
Rule: $H \leq 22$ (no learning)	fixed	1.000	0.220	0.361	n/a	n/a	189 / 669
A Benchmark (Random Forest)	0.50	0.698	0.354	0.470	0.643	0.875	132 / 241
B Kinematic (Logistic Reg.)	0.50	0.439	0.190	0.266	0.231	0.729	83 / 353
B Kinematic (Logistic Reg.)	tuned, thr 0.42	0.942	0.083	0.153	0.231	0.729	178 / 1,965
C Velocity + Sentry (Log. Reg.)	0.50	0.429	0.148	0.220	0.191	0.684	81 / 467
C Velocity + Sentry (Log. Reg.)	tuned, thr 0.42	0.921	0.076	0.141	0.191	0.684	174 / 2,102

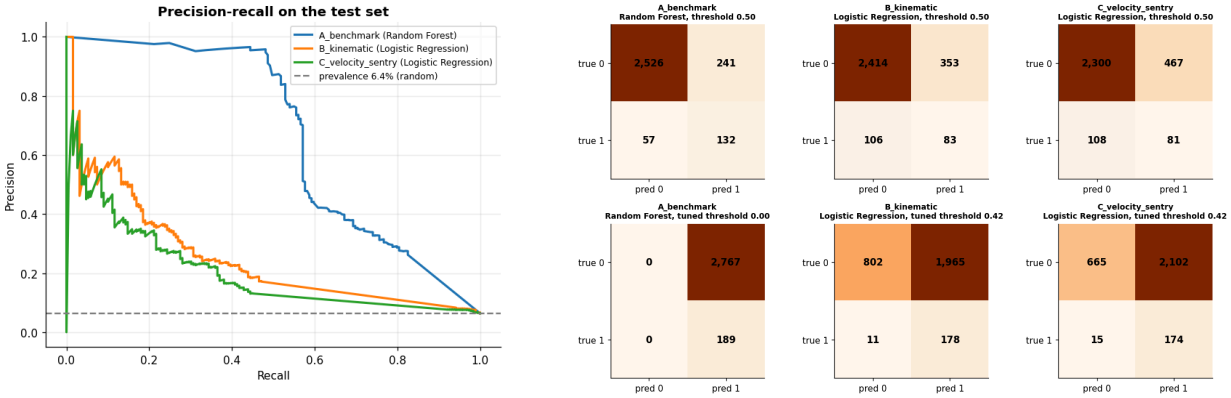


Figure 3. Precision-recall curves on the test set (left); confusion matrices at 0.50 and at the tuned threshold (right).

5.3 Interpretation, including where the models underperform

- **The benchmark is a ceiling, not a finding.** The one-line rule H ≤ 22 reaches recall 1.00 at precision 0.22, against the Random Forest's recall of 0.70 at precision 0.35. The model never sees MOID, so it recovers the size half of the definition but not the orbit half. For set A the tuned threshold reaches 90% recall only by flagging all 2,956 test asteroids (precision 0.064, equal to prevalence); this is a failure to separate classes, so the 0.50 result is the one to use.
- **Removing H and diameter costs a lot, but signal remains.** PR-AUC falls from 0.643 to 0.231 (set B), 3.6× the 6.4% prevalence and 3.7× the stratified-dummy baseline of 0.063. Much of that signal is the minimum miss distance acting as a proxy for MOID (Insight 3), not independent dynamics.
- **The velocity-only ablation is weak.** With every distance feature removed, PR-AUC is 0.191 (3.0× prevalence). Speed and Sentry membership describe orbit shape and discovery priority more than the PHA rule.
- **Trade-off.** Tuning for about 90% recall on sets B and C reaches 0.92-0.94 test recall (Wilson 95% interval for B: 0.90-0.97, n = 189) but precision drops to 0.08: roughly 11-12 false alarms per true hit. For hazard screening that cost may be acceptable because follow-up observation is cheap, but it is the price of recall and should not be hidden.
- **Random Forest vs Logistic Regression.** Random Forest wins only on set A (CV F1 0.46 vs 0.36). On B and C it is conservative at 0.50 (recall 0.12) and selection by CV F1 picks Logistic Regression. Permutation importance on set A is dominated by minimum miss distance and mean speed, not H, because H and the two diameters are near-duplicates and share importance.

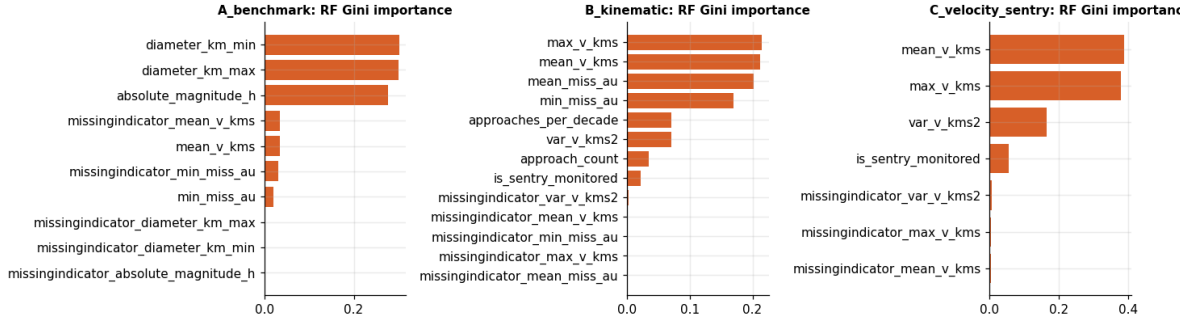


Figure 4. Random Forest Gini importance for the three feature sets (indicator columns flag missing approach statistics).

6. Dashboard walkthrough (Power BI)

The .pbix connects to the flat CSVs in the clean folder (Power BI Desktop reads SQLite only through an ODBC driver; the CSVs hold identical data). fact_close_approach, fact_sentry and ml_predictions each relate many-to-one to dim_asteroid on asteroid_id. The report has three pages and tells one story for a non-technical reader: what is out there, which objects deserve a closer look, and how far a model can take us.

Page	Visuals	Message and interaction
1. Overview	KPI cards (14,776 asteroids, 943 PHAs = 6.4%, 14,573 approaches, 783 Sentry-monitored); line chart of approaches per decade by orbit class; PHA rate by orbit class	Hazard is rare and concentrated; the rising approach count is a discovery effect (annotated). A decade slicer filters the page.
2. Drill-down	Asteroid table with size-class and orbit-class slicers; approach timeline; scatter of minimum miss distance vs MOID coloured by PHA flag	Pick any asteroid and drill through to its approach history. Size and orbit slicers support triage.
3. Model insights	Confusion matrices, precision-recall curve, feature importance, predicted vs actual from the ml_ tables; filter split = test	Shows the recall/precision trade-off and why the benchmark feature set is flattering. A feature-set slicer compares A, B and C.

7. Limitations and decisions that could bias conclusions

- **The label is a definition, not ground truth.** Models learn the H ≤ 22 / MOID ≤ 0.05 rule and its data-timing quirks (23 mismatches), not physical impact risk.
- **Size is modelled, not measured.** NeoWs diameters are derived from H with an assumed albedo range, so set A has built-in leakage; it is documented and used as a ceiling.
- **Distance features are bounded by MOID.** Even the kinematic set contains part of the MOID condition; set C is the cleanest test of dynamic information alone.
- **Sampling.** NeoWs is a seeded random sample of catalogue pages; a different seed or catalogue update shifts the numbers slightly. 44.7% of sampled asteroids have a CAD approach under 0.05 au, so distance statistics are censored for the rest.
- **Discovery bias.** Approach counts reflect observation history and projected passes, not true encounter rates (Insight 1). Objects outside the sample were dropped from fact tables to keep foreign keys valid; they stay in the raw snapshot.
- **Imputation is not neutral.** The 'no approach' indicator is itself informative, which is correct but means imputation shapes the model. **Statistical power:** 189 test positives give wide recall intervals, so small differences between models are unreliable. **Snapshots:** Sentry and NeoWs change as observations arrive; results describe the pull recorded in the manifest.

8. Conclusion

A hazardous label can be reproduced almost perfectly from size, because size is the definition. Without H and diameter, encounter behaviour still lifts PR-AUC about 3.6× above chance, but most of that is minimum miss distance standing in for MOID; speed and monitoring status alone carry only weak signal. The trade-off is explicit: pushing recall to 90-94% on kinematic features costs about eleven to twelve false alarms per true hazard. Future work: add MOID as an explicit feature and measure its marginal value, use a larger NeoWs sample, and model discovery effort so count features stop measuring observation history.